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SJB INSTITUTE OF TECHNOLOGY
(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)
Approved by AICTE, New Delhi, Recognized by UGC, New Delhi with 2 (f) & 12 (B)
Accredited by NAAC with 'A+' Grade, Accredited by National Board of Accreditation)
No. 67, BGS Health & Education City, Dr. Vishnuvardhan Road, Kengeri, Bengaluru-560060.

DEPARTMENT OF
ELECTRONICS & COMMUNICATION ENGINEERING

Formative Assessment -2 Report
VLSI Design & Testing – (23ECT601)

Name	Likith Kumar K
USN	1JB23EC048
Semester & Section	6^{th} A

Evaluation Rubrics:

Sl. No.	Criteria	Description	Max. Marks	Scored Marks
1.	**Topic Knowledge**	Depends on depth of understanding, clarity of concepts related to a particular field or topic. It indicates how well a student understands the fundamental principles, advanced concepts and applications.	10	
2.	**Innovation in Design**	Refers to the process of creating new, improved, or more efficient solutions through creative thinking, advanced technology, and practical problem-solving approaches	15	
3.	**Results & Conclusion**	Shows what was achieved and what it means.	15	
4.	**Report**	A structured and formal document that presents information, analysis, findings, and conclusions as per the format.	10	
		Total	**50**	

Student Signature

Staff Signature

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Formative Assessment -2 Report
VLSI Design & Testing – (23ECT601)

Name	**Harshith D M**
USN	**1JB23EC033**
Semester & Section	**6th A**

Evaluation Rubrics:

Sl. No.	Criteria	Description	Max. Marks	Scored Marks
1.	**Topic Knowledge**	Depends on depth of understanding, clarity of concepts related to a particular field or topic. It indicates how well a student understands the fundamental principles, advanced concepts and applications.	10	
2.	**Innovation in Design**	Refers to the process of creating new, improved, or more efficient solutions through creative thinking, advanced technology, and practical problem-solving approaches	15	
3.	**Results & Conclusion**	Shows what was achieved and what it means.	15	
4.	**Report**	A structured and formal document that presents information, analysis, findings, and conclusions as per the format.	10	
		Total	**50**	

Student Signature

Staff Signature

1. Introduction

Static Random Access Memory (SRAM) is a fundamental building block in modern digital integrated circuits. Unlike Dynamic RAM (DRAM), SRAM does not require periodic refresh cycles, making it faster and more suitable for on-chip memory applications such as cache memories in microprocessors and embedded systems.

The 6-transistor (6T) SRAM cell is the most widely adopted SRAM architecture in industry due to its robust stability, relatively compact area, and ease of integration in standard CMOS processes. It consists of two cross-coupled CMOS inverters that store the data bit, and two access (pass-gate) transistors controlled by a Word Line (WL) that allow read and write operations via complementary Bit Lines (BL and BLB).

This report presents the complete design, schematic implementation, symbol creation, testbench setup, and transient simulation results of a 6T SRAM cell implemented in the Cadence Virtuoso environment and simulated using the Spectre simulator.

2. 6T SRAM Cell Structure

2.1 Transistor Composition

The 6T SRAM cell consists of six MOSFETs arranged as follows:

- Two PMOS pull-up transistors (M1, M2) — form the load devices of the cross-coupled inverters
- Two NMOS pull-down transistors (M3, M4) — form the driver devices of the cross-coupled inverters
- Two NMOS access transistors (M5, M6) — gate controlled by the Word Line (WL) to connect the storage nodes to the Bit Lines (BL, BLB)

2.2 Storage Nodes

The internal storage nodes Q and QB are complementary. When Q = logic '1' (~VDD = 1.8 V), QB = logic '0' (~VSS = 0 V), and vice versa. The cross-coupled inverter pair provides positive feedback that latches and maintains the stored state indefinitely as long as power is supplied.

Figure 1: 6T SRAM Cell Schematic — Cadence Virtuoso Schematic Editor (analogggg library, 6TDRAM cell)

As seen in Figure 1, the schematic contains two cross-coupled inverters at the core and two pass-gate NMOS transistors connected to the BL and BLB ports. The WL signal controls both access transistors simultaneously. The transistors are sized to satisfy the read stability and write-ability constraints through proper cell ratio and pull-up ratio selection.

3. Cell Symbol

A custom symbol was created in the Cadence Symbol Editor to represent the 6T SRAM cell as an abstracted block for use in higher-level testbench schematics. The symbol exposes five pins: BL (Bit Line), BLB (Bit Line Bar), WL (Word Line), VDD (Power Supply), and VSS (Ground).

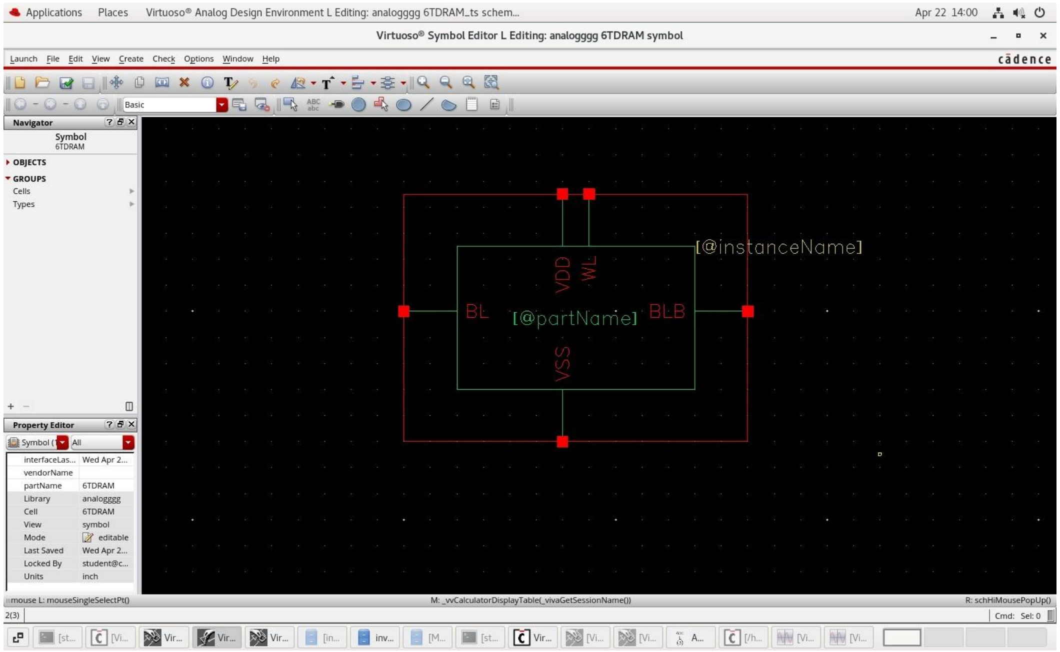

Figure 2: 6T SRAM Symbol — Cadence Virtuoso Symbol Editor (analogggg library, 6TDRAM symbol)

The symbol shown in Figure 2 uses the [@partName] and [@instanceName] annotations to display the cell name and instance identifier on the schematic canvas. The pin placement follows standard convention: BL on the left, BLB on the right, WL and VDD at the top, and VSS at the bottom, ensuring intuitive connectivity in testbench designs.

4. Testbench Setup

4.1 Testbench Schematic

A dedicated testbench schematic (6TDRAM_ts) was constructed in Cadence ADE L to drive and observe the 6T SRAM cell. The testbench provides voltage stimuli on all input ports and monitors the output voltages and currents for transient analysis.

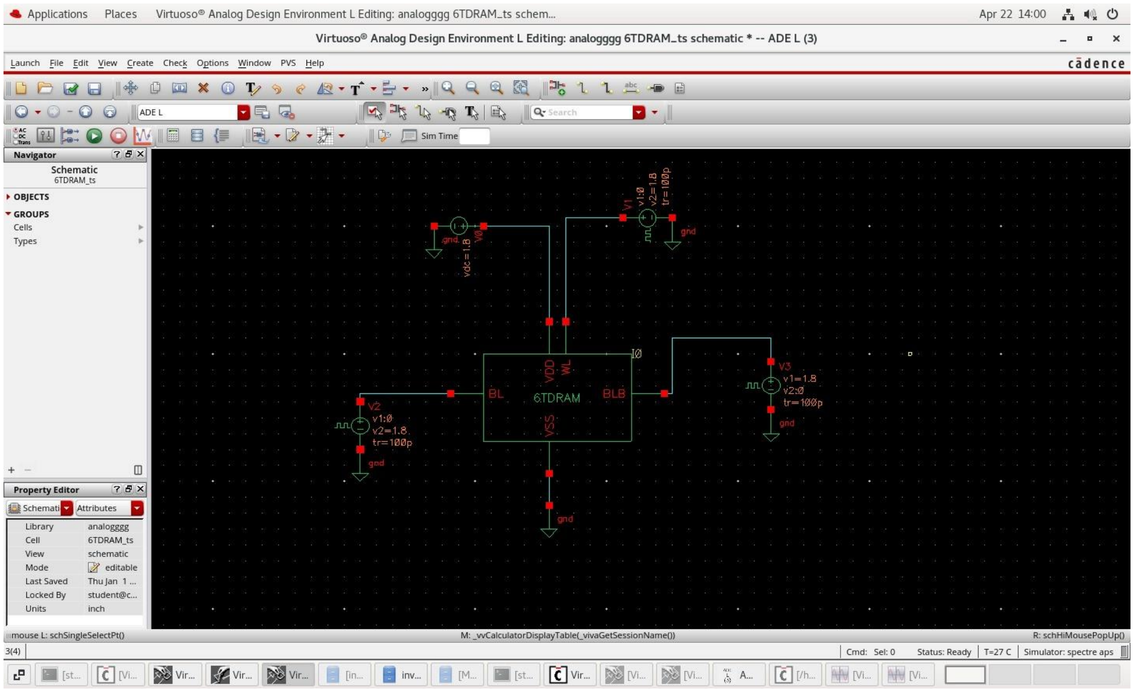

Figure 3: Testbench Schematic — 6TDRAM_ts in Cadence ADE L Environment

4.2 Stimulus Sources

The following voltage sources are used in the testbench (Figure 3):

- V1 — DC voltage source (VDC = 1.8 V) connected to VDD to supply the SRAM cell
- V2 — Pulse source on BL: v1 = 0 V, v2 = 1.8 V, tr = 100 ps (write stimulus for BL)
- V3 — Pulse source on BLB: v1 = 1.8 V, v2 = 0 V, tr = 100 ps (complementary write stimulus)
- WL — Pulse source controlling the word line timing to enable and disable the access transistors

The simulation is a transient analysis run over 20 ns, configured in the Cadence Analog Design Environment L (ADE L) with the Spectre simulator at a temperature of T = 27°C.

5. Simulation Results

5.1 Observed Waveforms

The transient simulation produces five waveforms observed in the Virtuoso Visualization & Analysis XL (VIVA) environment:

- /IO/BL — Current through the BL node (range: -1.16 to 0.706 mA)
- /IO/BLB — Current through the BLB node (range: -1.17 to 0.697 mA)

- /IO/WL — Current through the WL node (range: -139.5 to 174.5 uA)
- /IO/Q — Voltage at internal storage node Q (range: -0.189 to 1.99 V)
- /IO/QB — Voltage at complementary storage node QB (range: -0.188 to 1.99 V)

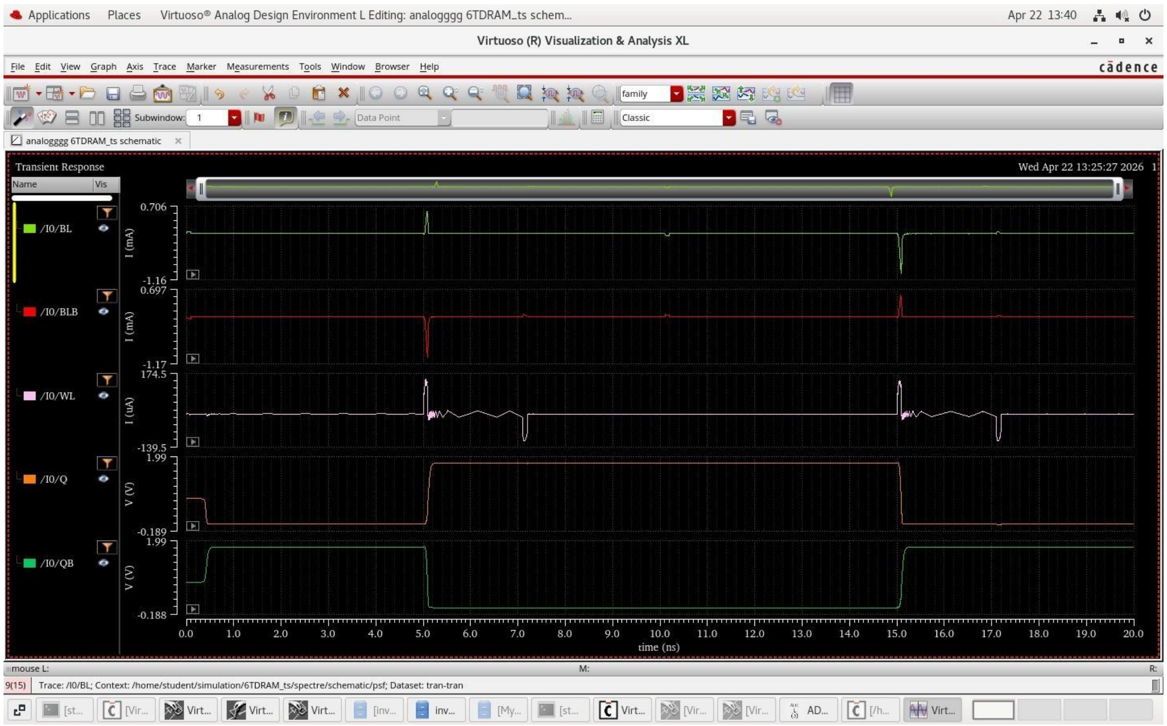

Figure 4: Transient Simulation Waveforms — BL, BLB, WL currents and Q, QB voltages over 20 ns

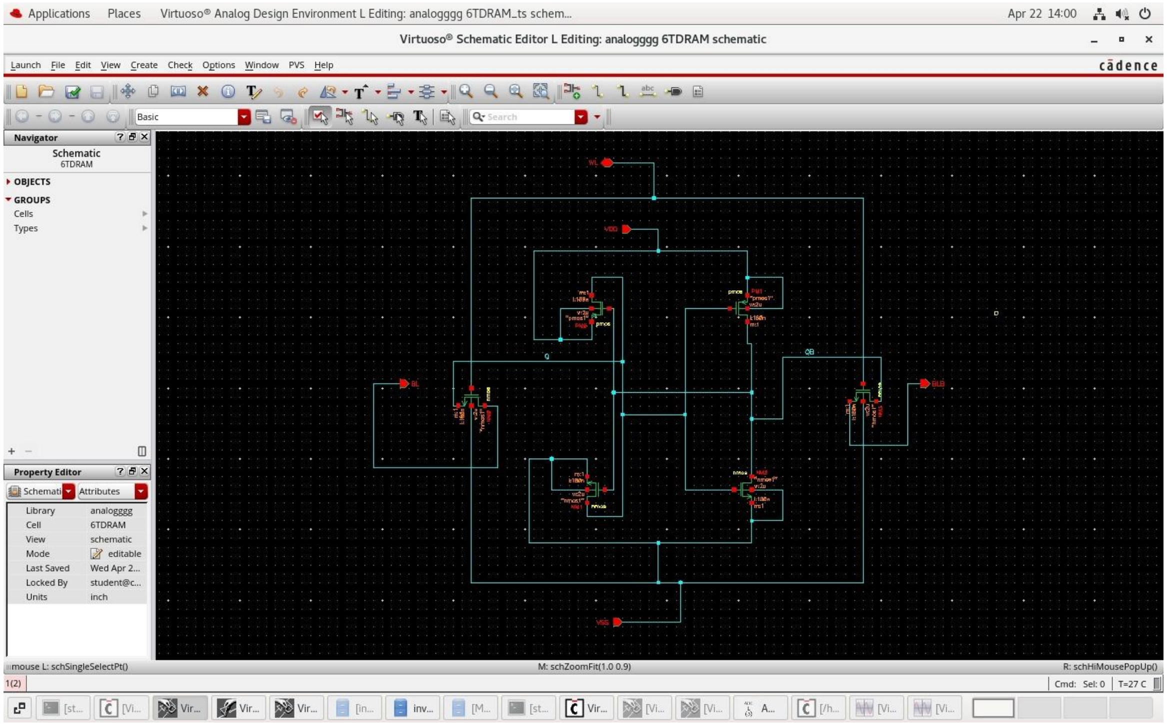

Figure 5: Transient Simulation Waveforms (Detailed View) — Full view of all five signal traces

5.2 Waveform Analysis

From the simulation waveforms (Figures 4 and 5), the following operational behavior is confirmed:

Initial State (0–5 ns): The SRAM cell is in a stable hold state. Q is near VSS (~0 V) and QB is near VDD (~1.8 V). The BL and BLB currents are negligible, confirming no access activity. The WL remains inactive (low), keeping the pass-gate transistors in the off state.

Write Operation (~5 ns): A write event is triggered when WL is pulsed HIGH, enabling both access transistors. BL is driven to 1.8 V and BLB to 0 V, which forces Q to transition from 0 V to ~1.8 V and QB from ~1.8 V to 0 V. Brief current spikes are visible on BL (positive, ~0.706 mA) and BLB (negative, ~-1.16 mA) during the transition — characteristic of the charge and discharge of the storage nodes through the access transistors.

Hold State After Write (~7–15 ns): After WL falls back to LOW, the cell retains the newly written state. Q remains stable at ~1.8 V and QB at ~0 V, confirming successful write and stable latch behavior with no data loss.

Second Write (~15 ns): A second write cycle near 15 ns flips the cell back to its original state. Q transitions from high to low and QB from low to high, demonstrating symmetric write capability in both directions. The WL current waveform shows small oscillatory spikes during word-line transitions, attributed to gate capacitance charging of the access transistors — this is expected and consistent with normal SRAM operation.

6. Design Specifications Summary

Table 1 below summarizes the key design and simulation parameters for the 6T SRAM cell implemented in this work:

Parameter	Value
Supply Voltage (VDD)	1.8 V
Technology Node	CMOS (180 nm / tsmc)
Cell Transistors	6 (2 PMOS + 4 NMOS)
Bit-line Swing	0 V to 1.8 V
Word Line Pulse	100 ps rise/fall time
Simulation Time	20 ns (transient)
Simulator	Cadence Spectre

Table 1: 6T SRAM Design and Simulation Parameters

7. Stability Considerations

7.1 Read Stability

During a read operation, both BL and BLB are pre-charged to VDD. When WL is asserted, the access transistors connect the bit lines to the storage nodes. The pull-down NMOS transistors must be stronger than the access transistors to prevent inadvertent flipping of the stored data. This condition is satisfied by maintaining a Cell Ratio (CR = W/L of pull-down / W/L of access) greater than unity.

7.2 Write Ability

During a write operation, one bit line is driven to the opposite logic level of the stored value. The access transistors must be capable of overpowering the pull-up PMOS transistor to flip the cell state. This is characterized by the Pull-up Ratio (PR = W/L of pull-up / W/L of access), which must be less than 1 to guarantee write-ability. The simulation confirms successful write operations in both directions, validating the transistor sizing choices.

7.3 Static Noise Margin (SNM)

The Static Noise Margin (SNM) quantifies the robustness of the SRAM cell against noise during hold and read operations. It is derived from the butterfly curve of the two cross-coupled inverter voltage transfer characteristics (VTCs). The maximum square inscribed within the butterfly lobes defines the SNM. A larger SNM indicates a more stable cell. In the 6T topology, the read SNM is typically smaller than the hold SNM due to the voltage divider formed by the access and pull-down transistors, making read stability the critical design constraint.

8. Conclusion

A complete 6-transistor SRAM cell was successfully designed, symbolized, and verified through transient simulation in the Cadence Virtuoso environment using the Spectre simulator. The simulation results confirm correct and stable write and hold operations at a supply voltage of 1.8 V over a 20 ns simulation window.

The cell demonstrates clean voltage transitions at Q and QB nodes with appropriate current activity on the bit lines during access events. Both write-to-high and write-to-low operations were verified, confirming the symmetry of the design. The WL current spikes during transitions are within expected bounds, indicating well-controlled gate capacitance effects.

The design fulfills the fundamental requirements of SRAM operation: reliable data retention in the hold state, correct write behavior when WL is asserted, and symmetric operation for writing both logic levels. Further characterization including SNM extraction, power estimation, and PVT (Process, Voltage, Temperature) corner simulations would complete the full verification flow.

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